Hints: Number theory – 7

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Theorem 1(a): clearly $\lfloor f(\lfloor x\rfloor)\rfloor \leqslant \lfloor f(x)\rfloor$. If the inequality were strict, there would be an integer k with $f(\lfloor x\rfloor) < k \leqslant f(x)$. By continuity $f(y) = k$ for some $y \in (\lfloor x\rfloor, x]$, but y is not an integer. For (b), follow the hint in the statement: the difference of the two sides has period $\frac{1}{n}$ and vanishes on $[0, \frac{1}{n})$.

Hint 2. Apply Hermite with $n = 2$, $y = n/2^{k+1}$: $\lfloor y + \frac{1}{2}\rfloor = \lfloor 2y\rfloor - \lfloor y\rfloor$. So $\lfloor \frac{n+2^k}{2^{k+1}}\rfloor = \lfloor n/2^k\rfloor - \lfloor n/2^{k+1}\rfloor$, and the sum telescopes to n.

Hint 3. Theorem 2: count lattice points in the open rectangle $(0,q)\times(0,p)$ on each side of the diagonal. There are none on the diagonal itself when $\gcd(p,q) = 1$, and the point reflection $(x,y) \mapsto (q-x, p-y)$ swaps the two halves.

Hint 4. General d: write $p = dp'$, $q = dq'$. Then $\lfloor kp/q\rfloor = \lfloor kp'/q'\rfloor$. Split $0 \leqslant k < q$ into d blocks of length q'; in block j, $\lfloor kp'/q'\rfloor$ increases by jp'. Sum: $d\cdot\frac{(p'-1)(q'-1)}{2} + p'q'\frac{d(d-1)}{2}$, which simplifies to the claimed formula.

Problem 2

Hint 1. Expand $x^{m-1}(1-x)^{n-m}$: it has integer coefficients and degree $n-1$, so $L(n)$ times its integral is an integer. Integrating by parts (or using the Beta function), the integral equals $\frac{(m-1)!(n-m)!}{n!} = \frac{1}{m\binom{n}{m}}$.

Hint 2. Nair: $k\binom{2k}{k} \mid L(2k) \mid L(2k+1)$ and $(2k+1)\binom{2k}{k} = (k+1)\binom{2k+1}{k+1} \mid L(2k+1)$. Since $\gcd(k, 2k+1) = 1$, $k(2k+1)\binom{2k}{k} \mid L(2k+1)$. With $\binom{2k}{k} \geqslant \frac{4^k}{2k+1}$ this gives $L(2k+1) \geqslant k4^k \geqslant 2^{2k+2}$ for $k \geqslant 4$.

Hint 3. Even n: $L(2k+2) \geqslant L(2k+1)$. Small cases: $L(7) = 420$, $L(8) = 840$.

Hint 4. Upper bound: every prime power in $(k, 2k]$ divides $\binom{2k}{k}$ (Legendre/Kummer), so $L(2k) \mid L(k)\binom{2k}{k}$. Similarly $L(2k+1) \mid L(k+1)\binom{2k+1}{k}$. Induct, using $\binom{2k}{k} \leqslant 4^k$ and $\binom{2k+1}{k} \leqslant 4^k$.

Problem 3

Hint 1. Even pq: then $p+q$ is odd. Use the involution $k \mapsto pq-1-k$, and show that $\lfloor \frac{pq-1-k}{p}\rfloor = q - 1 - \lfloor \frac{k}{p}\rfloor$.

Hint 2. So the exponent changes to $(p+q-2) - (\text{old exponent})$. This has the opposite parity, so the terms cancel in pairs and the sum is 0.

Hint 3. Odd pq: let $r = k \bmod p$, $s = k \bmod q$. Since p is odd, $\lfloor k/p\rfloor = \frac{k-r}{p} \equiv k - r \pmod 2$, and similarly for q. Hence the sign is $(-1)^{r+s}$.

Hint 4. By CRT, as k runs over $0, \ldots, pq-1$ the pair (r,s) runs over all of $\{0,\ldots,p-1\}\times\{0,\ldots,q-1\}$. So the sum factors as $\left(\sum_{r<p}(-1)^r\right)\left(\sum_{s<q}(-1)^s\right) = 1\cdot 1 = 1$.

Problem 4

Hint 1. Test $b = 1$ first. Since a is even, $1 + x \geqslant 0$ is needed, so $x \geqslant -1$. Also use Theorem 1(c): $\sqrt[a]{b^a + xb^{a-1}} \leqslant b + \frac{x}{a}$.

Hint 2. For $-1 \leqslant x < a$ the identity holds. If $0 \leqslant x < a$, the root lies in $[b, b+1)$. If $-1 \leqslant x < 0$, use $(b-1)^a \leqslant b^{a-1}(b-1) \leqslant b^a + xb^{a-1} < b^a$.

Hint 3. For $x \geqslant a$, put $m = \lfloor x/a \rfloor \geqslant 1$ and $x = am + r$, $0 \leqslant r < a$. Then $b = 1$ requires $1 + x \geqslant (1+m)^a$, i.e. $r \geqslant (1+m)^a - 1 - am \geqslant \binom{a}{2} m^2$. Since $r < a$, this forces $a = 2$, $m = 1$, $r \geqslant 1$.

Hint 4. For $a = 2$ and $x \in [3, 4)$, check $(b+1)^2 \leqslant b^2 + xb < (b+2)^2$ for all $b \geqslant 1$. Answer: $x \in [-1, a)$ for $a \geqslant 4$, and $x \in [-1, 2) \cup [3, 4)$ for $a = 2$.

Problem 5

Hint 1. Structure of $\tau \in T_n$: let C be the set of cyclic points ($|C| = m$) and L the lcm of the cycle lengths on C, so $L \leqslant f(m)$. Let d be the largest distance of a point from C. Show that $\tau^i = \tau^j$ if and only if $i, j \geqslant d$ and $i \equiv j \pmod{L}$.

Hint 2. Hence $\operatorname{ord}(\tau) = L + d - 1 \leqslant f(m) + (n - m)$. If $n - m < n^{0.501}$, this is already $< f(n) + n^{0.501}$, since f is non-decreasing.

Hint 3. Otherwise $n - m \geqslant n^{0.501}$. The number $f(m)$ has at most $\sqrt{2m}$ distinct prime factors, since they are distinct primes with sum $\leqslant m$. But there are $\gg n^{0.501}/\ln n > \sqrt{2n}$ primes below $n^{0.501}$, so some prime $p \leqslant n^{0.501} \leqslant n - m$ does not divide $f(m)$. Adding a p-cycle gives $f(n) \geqslant f(m+p) \geqslant pf(m) \geqslant 2f(m)$.

Hint 4. Finish: if $f(m) \geqslant n$, then $f(n) \geqslant f(m) + n > \operatorname{ord}(\tau)$. If $f(m) < n$, then $\operatorname{ord}(\tau) < 2n \leqslant f(n)$ for large n (two coprime cycles of length about $n/2$).